Homework rules

- Group discussion is *highly recommended*. But you have to write down your solution by *yourself*. Aim for clarify and conciseness in your solution, emphasizing the main ideas over low-level details.
- You should *cite* all the references that help you to solve the homework, including the people you have discussed with and the online material you looked at.
- You are required to type your solution using latex and submit a *pdf* to Gradescope and a *tex* file to NTUCool. You may type your solution in English (preferable) or Chinese.
- You are not expected to solve *all* problems. Just try your best! If you don't solve a problem, try your best to write down your thinking process.
- You may use AI however you wish to deepen your understanding of the lecture material. Upload the notes, talk to your AI about them, ask for more explanation or examples; it's all fine. You may not use LLMs in any way to work on your homework. You may not upload assignments, ask for hints, ask how certain concepts from the lectures might be applied to specific homework problems, or upload your assignments to check for correctness or clarity or anything else. You may not include any AI generated content whatsoever in your homework submissions (This policy is taken from MIT 6.5620).
- We will use a *random-checking policy* in which students may be asked to present their homework solutions during TA office hours, with the expectation that everyone will present *at least once*. Evaluation is based on *consistency*, meaning students must be able to reproduce the solutions they originally submitted. If a student fails this consistency check, they will receive a significant *percentage deduction* from their homework score.
- 作業總分為38分，可得最大計分為32分。
- Deadline 5/10(日)23:59，遲交時間最晚5/12(二) 23:59 (遲交成績打9折)，超過則以0分計算。

1. (6 points) **(Statistical distance)** Given two distributions $(\mathcal{X}, \mathcal{Y})$ over a finite set Ω, their statistical distance (also known as variational or L_1 distance) is defined as

$$\Delta(\mathcal{X}, \mathcal{Y}) \coloneqq \frac{1}{2} \sum_{\omega \in \Omega} |\mathcal{X}(\omega) - \mathcal{Y}(\omega)|.$$

Where $\mathcal{X}(\omega) \coloneqq \Pr[x \leftarrow \mathcal{X} : x = \omega]$ is the probability of getting output ω when sampling from $\mathcal{X}$.

(i) (2 points) Show that the following is an equivalent definition:

$$\Delta(\mathcal{X}, \mathcal{Y}) \coloneqq \sup_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)| = \max_{A \subseteq \Omega} |\mathcal{X}(A) - \mathcal{Y}(A)|,$$

where $\mathcal{X}(A)$ denotes the probability of $\mathcal{X}$ to be in A, and similarly for $\mathcal{Y}(A)$. The second equation holds because we only consider Ω being a finite set in this problem.

(ii) (2 points) Show that, if $X \approx_{t,\epsilon} Y$ holds for *all* t, then $\Delta(\mathcal{X}, \mathcal{Y}) \leq \epsilon$. In this case, we write $\mathcal{X} \simeq_s \mathcal{Y}$.

(iii) (2 points) Show that for two distribution $\mathcal{X}, \mathcal{Y}$ where $\Delta(\mathcal{X}, \mathcal{Y}) < \epsilon$, $X \approx_{t,\epsilon} Y$ holds for *any* t.

2. (8 points) **(Leftover Hash Lemma)** In this problem, you are asked to prove the leftover hash lemma you saw in the class. Recall the statement of the lemma as follows.

Theorem. *Let $n, m, q \in \mathbb{N}$ where q is a prime, and $\epsilon \in (0, 1)$ be parameters satisfying $m \geq n \log q + 2\log(1/\epsilon) + 1$. Let $A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$ be a uniformly random matrix over $\mathbb{Z}_q^{m \times n}$ and $r \xleftarrow{\$} \{0, 1\}^m$. Then the distribution $(A, r^T A)$ is ϵ-close to uniform. Namely, let u denote the uniform distribution over $\mathbb{Z}_q^n$, we have*

$$\Delta((A, r^T A), (A, u)) \leq \epsilon$$

Before proving the lemma, we discuss the interpretation of the lemma. Intuitively, $r \xleftarrow{\$} \{0,1\}^m$ has m-bit of "entropy", and $u \xleftarrow{\$} \mathbb{Z}_q^n$ has $n \log q$ bits of "entropy". The lemma says that the uniformly random matrix $A \xleftarrow{\$} \mathbb{Z}_q^{m\times n}$ can be viewed as a *randomness extractor* that extracts the randomness of r and turn it into a close-to-uniform random variable $r^T A$, provided that r has sufficient entropy (i.e., m is sufficiently large). Now, you are asked to prove this lemma in the following three steps.

(i) (2 points) Show that for any $x, y \in \mathbb{Z}_q^m$ such that $x \neq y$,

$$\Pr_{A \xleftarrow{\$} \mathbb{Z}_q^{m\times n}} [x^T A = y^T A] \leq 1/q^n.$$

(ii) (2 points) For a discrete random variable X, define the *collision probability* of X to be the probability that two independent samples of X taking the same value (i.e., they collide). Namely, define $cp(X) := \Pr[X = X'] = \sum_x \Pr[X = x]^2$, where X' denotes an independent copy of X. Use (i) to prove the following upper bound on the collision probability of $(A, r^T A)$:

$$cp(A, r^T A) \leq \frac{1}{q^{mn}} \cdot \left(\frac{1}{2^m} + \frac{1}{q^n}\right).$$

(iii) (2 points) Let X be a random variable with support size N. It is known that if X has collision probability $cp(X) \leq (1+\epsilon^2)/N$, then X is ϵ-close to the uniform distribution over its support. Use this fact and (ii) to prove the lemma.

(iv) (2 points) Prove the above stated fact, i.e., if X has collision probability $cp(X) \leq (1+\epsilon^2)/N$, then X is ϵ-close to uniform.

3. (8 points)(Fun with the Equivalent CPA definitions) Show that the following security definitions for PKE (defined by the following security games) are equivalent to CPA-security or not. Prove your answer or construct a counter-example.

$\mathsf{PKE}^{IND-CPA}$

$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$
$b \xleftarrow{\$} \{0,1\}$
$m_0, m_1 \leftarrow \mathbf{Adv}(\mathsf{pk})$
$b' \leftarrow \mathbf{Adv}(\mathsf{pk}, \mathsf{Enc}_{\mathsf{pk}}(m_b))$
return $b' = b$

(i) (2 points) $\mathsf{PKE}^{ZeroEnc}$	(ii) (2 points) $\mathsf{PKE}^{VectorEnc}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$ $b \xleftarrow{\$} \{0,1\}$ $m \leftarrow \mathbf{Adv}(\mathsf{pk})$ if $b = 0, ct = \mathsf{Enc}_{\mathsf{pk}}(m)$ otherwise, $ct = \mathsf{Enc}_{\mathsf{pk}}(0^{\lvert\mathcal{M}\rvert})$ $b' \leftarrow \mathbf{Adv}(\mathsf{pk}, ct)$ return $b' = b$	$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$ $b \xleftarrow{\$} \{0,1\}$ $(m_{j,0}, m_{j,1})_{j=1}^n \leftarrow \mathbf{Adv}(\mathsf{pk})$ $(ct_{j,b})_{j=1}^n = (\mathsf{Enc}_{\mathsf{pk}}(m_{j,b}))_{j=1}^n$ $b' \leftarrow \mathbf{Adv}(\mathsf{pk}, (ct_{j,b})_{j=1}^n)$ return $b' = b$

(iii) (2 points) $\mathsf{PKE}^{Same-or-not}$	(iv) (2 points) $\mathsf{PKE}^{Rand-Same-or-not}$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$ $b \xleftarrow{\$} \{0,1\}, r \xleftarrow{\$} \{0,1\}$ $m_0, m_1 \leftarrow \mathbf{Adv}(\mathsf{pk})$ if $b = 0$, $ct = \{\mathsf{Enc}_{\mathsf{pk}}(m_r), \mathsf{Enc}_{\mathsf{pk}}(m_r)\}$ otherwise, $ct = \{\mathsf{Enc}_{\mathsf{pk}}(m_r), \mathsf{Enc}_{\mathsf{pk}}(m_{\bar{r}})\}$ $b' \leftarrow \mathbf{Adv}(\mathsf{pk}, ct)$ return $b' = b$	$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^n)$ $b \xleftarrow{\$} \{0,1\}, m, m' \xleftarrow{\$} \mathcal{M}_n$ if $b = 0$, $ct = \{\mathsf{Enc}_{\mathsf{pk}}(m), \mathsf{Enc}_{\mathsf{pk}}(m)\}$ otherwise, $ct = \{\mathsf{Enc}_{\mathsf{pk}}(m), \mathsf{Enc}_{\mathsf{pk}}(m')\}$ $b' \leftarrow \mathbf{Adv}(\mathsf{pk}, ct)$ return $b' = b$

4. (5 points) **(CPA-secure Symmetric-Key Encryption (SKE) from PRF)**

Definition. A symmetric-key encryption scheme (SKE scheme) is a triple $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ of PPT algorithms with the following syntax:

- $\mathsf{KeyGen}(1^n)$ takes as input the security parameter 1^n and outputs a key k.
- $\mathsf{Enc}_k(m)$ takes as inputs a key k and a message $m \in \mathcal{M}_n$, where $\mathcal{M}_n$ is the space of messages, and outputs a ciphertext $ct \in \mathcal{C}_n$.
- $\mathsf{Dec}_k(ct)$ takes as inputs a key k and a ciphertext ct and outputs a message m or $\perp$ denoting failure.

It is required to satisfy the following correctness property:

- (Correctness) There exists a negligible function $\mu(\cdot)$ such that for every $n \in \mathbb{N}$ and every message $m \in \mathcal{M}_n$,

$$\Pr[k \leftarrow \mathsf{KeyGen} : \mathsf{Dec}_k(\mathsf{Enc}_k(m)) = m] \geq 1 - \mu(n).$$

Consider the following CPA experiment SKE^{CPA}:

- $k \leftarrow \mathsf{KeyGen}(1^n)$;
- The adversary $\mathbf{Adv}$ is given as input 1^n and the oracle access $\mathsf{Enc}_k(\cdot)$, and outputs a pair of messages m_0, m_1;
- A bit $b \xleftarrow{\$} \{0,1\}$ is uniformly chosen at random and a ciphertext $ct \leftarrow \mathsf{Enc}_k(m_b)$ is computed and is given to the adversary $\mathbf{Adv}$;
- The adversary $\mathbf{Adv}$ continues to have oracle access to $\mathsf{Enc}_k(\cdot)$ and outputs a bit b';
- $\mathbf{Adv}$ wins iff $b' = b$.

As in the PKE case, an SKE scheme $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ is *CPA-secure* if for all PPT adversaries $\mathbf{Adv}$ there is a negligible function $\mu(\cdot)$ such that, for any $n \in \mathbb{N}$, in the CPA experiment SKE^{CPA} we have

$$\Pr[\mathbf{Adv} \textit{ wins }] \leq \frac{1}{2} + \mu(n)$$

where the probability is taken over the randomness used by $\mathbf{Adv}$ as well as used in the experiment SKE^{CPA}.

Now, consider the PRF-based SKE scheme constructed from a PRF $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$, defined as following : (general key length/output length also works, but we here just consider the simple case.)

- $\mathsf{KeyGen}(1^n)$: sample a uniformly random $k \in \{0,1\}^n$.
- $\mathsf{Enc}_k(m)$: choose a uniformly random $r \in \{0,1\}^n$, and output $ct = (r, F_k(r) \oplus m)$.
- $\mathsf{Dec}_k((r,t))$: output $F_k(r) \oplus t$.

(a) (2 points) Show that this scheme satisfies the correctness condition.

(b) (3 points) Show that this scheme is CPA-secure.

5. (4 points) **(CCA-security)**

   (i) (2 points) Prove that El Gamal PKE is not IND-CCA-secure.

   (ii) (2 points) Prove that Regev's PKE is not IND-CCA-secure.

6. (7 points) **(Complete the proof in the class)** Recall that in the proof of Regev's PKE scheme being IND-CPA secure we first deduced from DLWE assumption that

$$(A, b_{\mathsf{pk}}, M_0, M_1, r^T A, r^T b_{\mathsf{pk}} + M_b \cdot \lfloor q/2 \rfloor) \simeq_c (A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \lfloor q/2 \rfloor). \qquad (1)$$

   (a) (3 points) Formally prove the equation (1) by constructing a reduction.

   Then we applied the Leftover Hash Lemma to show that

$$(A, u, M_0, M_1, r^T A, r^T u + M_b \cdot \lfloor q/2 \rfloor) \simeq_s (A, u, M_0, M_1, u_1, u_2). \qquad (2)$$

   (b) (4 points) Formally prove the equation (2) by constructing a reduction.